



In addition to part I (General Handout for all courses, appended to the Time table), this portion provides further specific details about the course.

Course No : **CHE F212**
Course Title : **Fluid Mechanics**
Instructor-in-Charge : **DR PRATIK N SHETH**
Tutorial Instructors : **Dr Pratik N Sheth, Dr Suresh Gupta**

1. Course Description:

Fundamental Concepts, Fluid Statics, Integral and Differential Analysis for Fluid Motion, Dimensional Analysis, Internal and External Fluid Flow, Fluid Machinery, Flow through Packed Bed, Agitation, Introduction to Compressible Flow.

2. Scope and Objective of the Course:

This course is an introduction to the field of fluid mechanics. It mainly covers the basic principles of fluid mechanics. It introduces the student to the fundamental and practical aspects of fluid flow operations that a practising chemical engineer encounters regularly. The physical concepts of fluid mechanics and analysis methods, beginning from basic principles, shall be dealt with in this course.

3. Learning Outcomes

- Recognize and define key fluid properties (such as pressure, density, temperature, viscosity, and velocity fields), calculate shear stresses and viscous forces, and apply fundamental fluid laws including the ideal gas law, surface tension, and speed of sound.
- Evaluate pressure distributions in hydrostatic fluids and submerged surfaces across various geometries, and predict buoyancy forces on floating bodies.
- Apply the Reynolds transport theorem to establish conservation laws and solve problems involving the Conservation of mass and momentum using integral and differential analysis.
- Apply Bernoulli and Euler equations to solve incompressible inviscid flow problems, and develop key dimensionless groups using principles like the Buckingham Pi Theorem.
- Analyze fully developed laminar and turbulent pipe flows, calculate pipe losses using the Moody chart, and evaluate pressure drops across fittings, valves, and sudden expansions or contractions.
- Apply boundary-layer concepts (including calculating thickness and pressure gradients), evaluate Drag on immersed bodies, analyze fluid flow through solid beds, and determine minimum fluidization velocities.
- Understand the characteristics and operations of pipes, fittings, and valves, and select and design pumps based on specific process requirements.

4. Text Books:

- T1 Fox, R.W. and A.T. McDonalds, Introduction to Fluid Mechanics (8th Ed.), John Wiley & Sons Inc., 2011.
- T2 McCabe, W.L., J.C. Smith and P. Harriott, Unit Operations of Chemical Engineering (7th Ed.), McGraw-Hill Inc., 2005.



5. Reference Books:

R1 Bird, R.B., W.E. Stewart and E.N. Lightfoot, Transport Phenomena, Revised 2nd edition (An Indian Adaptation), John Wiley and Sons Inc., 2021.

6. Course Plan:

Module No. and Title	Lecture Session	Reference	Learning outcomes
1. Introduction to Fluid Mechanics	L 1-2 Definition of a fluid, Basic equations, Dimensions and units, Dimensionless equations and Consistent units, Dimensional equations, Method of analysis.	TB1: Chap. 1 TB2: Chap.1	Recognize the meanings and definitions of different fluid properties, such as pressure, density, temperature, viscosity, and velocity fields.
2. Fundamental concepts of fluid mechanics	L 3-5 Concept of fluid, continuum, Velocity and stress field, Viscosity, Viscosity of gases and liquids, Surface tension, Description and classification of fluid motions.	TB1: Chap. 2	- Calculate shear stresses and viscous forces under various circumstances. - Understand the laws related to fluid properties, including the ideal gas law, surface tension, and the speed of sound.
3. Fluid Statics (Study of the principles of fluid statics and their application for various purposes)	L 6-8 Basic equations of fluid statics, Pressure variation in static fluids, Hydrostatic Equilibrium in a centrifugal field, Buoyancy and stability.	TB1: Chap. 3 TB2: Chap. 2	- Evaluate pressure distribution in a hydrostatic fluid and submerged surfaces for different geometries, including flat and curved surfaces. - Predict the buoyancy forces on the floating bodies.
4. Basic equations in integral form for a control volume	L 9-12 Basic laws for a system, Conservation of mass and momentum equations for integral control volumes,	TB1: 4.1-4.4	Apply the Reynolds transport theorem to obtain conservation laws of fluid mechanics.
5. Introduction to differential analysis of fluid motion	L 13-16 Conservation of mass and momentum equations	TB1: 5.1, 5.4	Solve the problems of Conservation of mass and momentum
6. Fundamentals of incompressible inviscid flow	L 17-19 Euler's equations, Bernoulli's equation,	TB1: 6.1-6.5	Apply the Bernoulli equation to solve flow problems
7. Dimensional Analysis and Similitude	L 20-22 Nondimensionalization, Buckingham Pi Theorem,	TB1: chap. 7	Development of significant dimensionless groups in fluid mechanics



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8. Internal incompressible viscous flow	L 23-29 Fully developed laminar flow, Flow in pipes and ducts, Flow measurements	TB1: Chap.8 TB2:Chap. 5 & 8	- Analyze fully developed laminar and turbulent pipe flows. - Calculate pipe losses using the Moody chart. - Evaluate pressure losses in fittings, valves and sudden enlargements/contractions.
9. External incompressible viscous flow	L 30-35 Boundary Layers, Fluid flow about immersed bodies, Boundary layer concept, Boundary layer thickness, Pressure gradient in the boundary layer, Drag & flow through beds of solids, fluidized beds, Sedimentation.	TB1: Chap.9 TB2: Chap. 7	- Apply the concept of the boundary layer. - Calculate boundary layer thickness and pressure gradients. - Evaluate drag in a bed of solids. - Analyze the characteristics of fluidized beds. - Determine minimum fluidization velocity.
10 Sedimentation, Transportation and metering of fluids	L 36-42 Pipe, Fittings, and Valves; Pumps: Classification & selection, Performance Characteristics	TB1: 10.3 TB2: Chap. 8	- Understand the functioning and characteristics of Pipes, fittings and valves. - Design and select pumps based on process requirements.

7. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 Mins	30	Oct 7, 2026 FN1	Closed Book
Comprehensive Examination	3 hours	40	Dec 8, 2026 FN	Closed Book + Open Book
Tutorial Tests* ^{\$}	10-15 minutes	20	During the tutorial class hour	Open Book/Closed Book
Assignment/Class Activity# ^{\$}	20 – 30 minutes	10	During lecture or tutorial hours	Based on the analysis of case-study/ design/ experiment /video/ story (individual/group)

* Best 6 tut marks out of 7 will be considered.

Best X assignment marks out of (X+1) will be considered.

\$ All tutorial tests would be announced, and all assignments would be surprise-based.

Midsem Grading would be based on the midsem test performance and the marks from three tutorial tests and two assignments.

8. Consultation Hour: Monday, 5 pm - 6 pm. The prior email with details of queries is encouraged.

9. Notices: would be displayed on Nalanda LMS.



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10. Makeup Policy: Makeup is granted only for genuine cases with valid justification and prior permission of the Instructor-in-charge. There is no makeup for the tutorial test or quiz.

11. NC Criteria: The minimum cut-off mark is at least 20% of the total marks to obtain a valid grade.

12. Attendance Policy: Each student is expected to attend all classes and participate in all evaluation components.

Instructor-in-charge
Course No. CHE F212